

El Moustpha Cheikh JIDDOU

Data & AI Engineer – Production Pipelines, RAG/LLM Systems and MLOps

@elmoustapha.cheikh.jiddou@gmail.com

+22248647376

Nouakchott, Mauritania



El Moustpha Cheikh Jiddou



MoustaphaCheikhJidou



Portfolio

PROFESSIONAL EXPERIENCE

Data Systems & IT Architecture Intern – Banque Nationale de Mauritanie (BNM) February 2025 – Present

 Nouakchott, Mauritania

- Designed and implemented a centralized enterprise Data Warehouse backed by robust ELT pipelines, automated flow orchestration, and predictive models.
- Architected and deployed a containerized microservices infrastructure over isolated Docker networks on on-premise production servers.
- Implemented a RAG (Retrieval-Augmented Generation) architecture with PostgreSQL/pgvector to index and query official banking documentation without data leakage.
- Benchmarked several LLMs (GPT-4o, Claude 3.5, Gemini 1.5, local Llama) to balance performance, infrastructure cost, and strict confidentiality.
- Developed sequential neural network models (LSTM, GRU) to anticipate rapid photovoltaic power drops and plan for critical infrastructure resilience.
- Environment:** Python (FastAPI), Data Warehousing, ELT pipelines, Docker, N8N, PostgreSQL (pgvector), PyTorch, React, REST APIs, Linux/Bash.

Assistant Engineer Intern – Mining & Geotechnical Department – Kinross Gold Corporation (Tasiast) July 2025 – September 2025

 Tasiast, Mauritania

- PV short-term forecasting: prepared minute-level data (lags, ramps ΔP , time encodings) and prototyped sequential models (LSTM/GRU) to anticipate power drops.

End-of-Study Internship – Caisse des Dépôts et de Développement (CDD) February – June 2023

 Nouakchott, Mauritania

- Developed computational solutions for risk analysis (logistic regression, k-NN, Random Forest) combining econometric models and machine learning.

PROJECTS

RAG-WebSystem – Generative AI system based on Retrieval-Augmented Generation

- End-to-end RAG engine: FAISS-based semantic search over indexed documents, contextual answer generation with an open-source model, exposed via a FastAPI backend and a React + Tailwind interface. GitHub.

Grain Size Analysis of 316L Microstructure – Computer-vision pipeline for grain boundary analysis

- Computer-vision pipeline segmenting grain boundaries in 316L stainless steel micrographs (500 $\times$), comparing thresholding, gradient filters, HED, and a U-Net model, with a Gradio interface. GitHub.

Poverty & Inequality Forecasting in Mauritania – Poverty indicator forecasts

- Forecasts four key poverty indicators from 1996 to 2030 using survey data and interpolation techniques. GitHub.

EDUCATION

Higher Polytechnic School of Nouakchott – ESP - Mauritania

 October 2023 – June 2026

State Engineer in Statistics and Data Engineering.

Faculty of Economic Sciences and Management – FSEG - Mauritania

 October 2023 – June 2025

Master's in Applied Econometrics and Statistics.

SKILLS

Python • FastAPI • Docker • PostgreSQL • PyTorch • n8n • Data Warehousing • ELT • RAG/LLM

LANGUAGES

Arabic 
Native Language

French 
DALF C1

English 
CEFR B1

VOLUNTEERING

President, ESP Data Club (January 2023 – April 2025)

CERTIFICATIONS

Google Advanced Data Analytics • freeCodeCamp Relational Database • DataCamp Data Engineering